

UNIVERSITY OF MARYLAND, BALTIMORE COUNTY

CMSC 481: Computer Networks

Fall 2025

Lectures:	Monday and Wednesday 2:30 PM -3:45 PM
Location:	Janet & Walter Sondheim 206
Instructor:	Dr. Yiwen Hu <huyiwen@umbc.edu >
TA:	TBD
Grader:	TBD
Prerequisite:	CMSC 341 or CMSC 341H with a grade of 'C' or better

Course Description

Networks are fundamental to modern life, enabling activities such as web browsing, email communication, video conferencing, online banking, e-commerce, and the management of smart home devices. Supporting these services requires complex computer networks that integrate a wide range of concepts, protocols, and technologies. The objective of this course is to provide students with a solid understanding of the principles underlying the design and operation of computer networks.

The course follows a top-down structure, starting with the application layer and progressing down to the physical layer. Key topics include network architectures and models, the World Wide Web and HTTP, transport-layer protocols (TCP and UDP), the Internet Protocol (IP) and routing, data link protocols, and an introduction to wireless networks such as Wi-Fi and 5G.

Office Hours

Instructor:	Dr. Yiwen Hu
<i>Location:</i>	ITE 322-A
<i>Time:</i>	Monday & Wednesday: 3:45 PM - 4:30 PM
<i>Email:</i>	huyiwen@umbc.edu

Teaching Assistant:	Md Nazmus Sakib
<i>Location:</i>	ITE 339
<i>Time:</i>	Tuesday 12:00 PM - 1:00 PM
<i>Email:</i>	msakib1@umbc.edu

Textbook and Reading Materials

The textbook below is highly recommended but not required. Additional reading materials may be shared during lectures to spark your interest and deepen your understanding.

Title: *Computer Networking: A Top-Down Approach*

Authors: James F. Kurose and Keith W. Ross

Edition: Latest edition (9th edition)

Publisher: Pearson

Communications

I encourage all students to engage actively in classroom discussions. I value your feedback throughout the semester. You are welcome to share your thoughts during office hours or reach out to me by email.

- All course announcements, materials, and assignment submissions will be handled through **Blackboard**.
- **Blackboard discussion board** is a place for general questions or clarifications. Please do not share solutions or post materials that directly reveal answers.
- You can also **email** us for questions or assistance. When you do, please start your subject line with [CMSC481] and send your email to BOTH the instructor and TA using your UMBC email.
 - We'll do our best to reply within 24 hours on weekdays. Emails that don't follow instructions or are sent at the last minute (e.g., right before an assignment is due) may not receive a response, so please plan ahead.

Grading

This section highlights the grading breakdown, grading scale, and important details about exams, homework, projects, and quizzes.

- **Grading Composition:** The course grade will be based on your performance in assignments, projects, quizzes, and exams as follows:
 - Midterm exam: 20%
 - Final exam: 25%
 - Homework assignments: 30%
 - Project: 20%
 - In-class quizzes (no prior announcements): 5%

- **Grading Scale:** Grades are based on the percentage of total points earned (X). The cutoffs below may be adjusted depending on overall class performance.

- $90\% \leq X$: A
- $80\% \leq X < 90\%$: B
- $70\% \leq X < 80\%$: C
- $60\% \leq X < 70\%$: D
- $X < 60\%$: F

- **Exams:** There will be one midterm and one final exam. Details such as locations, time slots, and required materials will be announced in class. No makeup exams will be allowed except in extreme circumstances with valid, documented proof of absence.
- **Homework Assignments and Project:** There will be three or four homework assignments, including conceptual questions, exercises, and tasks using network tools. There will also be one programming project. All homework assignments and the programming project are individual tasks and must be completed by each student independently.

The due dates for assignments and the project will be announced on Blackboard, with a deadline of 11:59 PM (UMBC time) on the specified day, unless stated otherwise. Submission instructions will be provided as well.

Late Policy: Late assignments and project will be accepted with the following penalties applied to the earned score:

- Up to 12 hours late: **-10%**
- 12–24 hours late: **-20%**
- 24–48 hours late: **-40%**
- More than 48 hours late: **no credit**

- **Quizzes:** There will be five quizzes during class, each consisting of 2–3 questions. Quizzes will not be announced in advance, and no makeup quizzes will be allowed.

Lecture Notes (Tentative)

Lecture	Date	Subject
L01	August 27, 2025	Course introduction and overview
L02	September 3, 2025	Network performance: lost, delay, throughput
L03	September 8, 2025	Protocol layers, service models
L04	September 10, 2025	Network layer: router, IPv4, addressing
L05	September 15, 2025	Network layer: NAT, IPv6, tools
L06	September 17, 2025	Network layer: forwarding, SDN
L07	September 22, 2025	Network layer: routing
L08	September 24, 2025	Network layer: routing
L09	September 29, 2025	Network layer: ICMP, network management
L10	October 1, 2025	Application layer: architecture, service requirements
L11	October 6, 2025	Application layer: Web and HTTP
L12	October 8, 2025	Application layer: socket programming, email
L13	October 13, 2025	Application layer: socket programming, course project
L14	October 15, 2025	Application layer: DNS, video streaming, CDNs
L15	October 20, 2025	Review
L16	October 22, 2025	Midterm
L17	October 27, 2025	Transport layer: (de)multiplexing, UDP
L18	October 29, 2025	Transport layer: Go Back N, Selective Repeat, TCP
L19	November 3, 2025	Transport layer: TCP, reliable data transfer
L20	November 5, 2025	Transport layer: flow control, congestion control
L21	November 10, 2025	Transport layer: congestion control
L22	November 12, 2025	Link layer: overview, error detection and correction
L23	November 17, 2025	Link layer: multiple access links and protocols
L24	November 19, 2025	Link layer: switched local area networks
L25	November 24, 2025	Wireless network: Wi-Fi, Bluetooth, satellite
/	November 26, 2025	No Class
L26	December 1, 2025	Wireless network: cellular 4G/5G
L27	December 3, 2025	Wireless network: mobility
L28	December 8, 2025	Review and Summary
/	December 12, 2025	Final exam (1:00-3:00 PM)

Academic Integrity

This course adheres to the academic honesty policy outlined at <https://academicconduct.umbc.edu/>. The following is a brief summary of the policy:

“Academic integrity is an important value at UMBC. By enrolling in this course, each student assumes the responsibilities of an active participant in UMBC’s scholarly community in which everyone’s academic work and behavior are held to the highest standards of honesty. Cheating, fabrication, plagiarism, and helping others to commit these acts are

all forms of academic dishonesty, and they are wrong. Academic misconduct could result in disciplinary action that may include, but is not limited to, suspension or dismissal.”

Please complete your own work and do not seek inappropriate assistance from others. If you are struggling with the course content, it is your responsibility to communicate with the instructor and/or TA so that we can provide assistance.

If you use ChatGPT, similar AI tools, or reference code or solutions found online, you must fully disclose how they were used. Failure to disclose is considered an academic integrity violation. Properly disclosed use may receive partial or full credit, depending on the extent of edits, achievement of learning outcomes, and overall impact on the assignment.

UMBC Policies

Accessibility and Disability Accommodations, Guidance and Resources

Accommodations for students with disabilities are provided for all students with a qualified disability under the Americans with Disabilities Act (ADA & ADAAA) and Section 504 of the Rehabilitation Act who request and are eligible for accommodations. The Office of Student Disability Services (SDS) is the UMBC department designated to coordinate accommodations that creates equal access for students when barriers to participation exist in University courses, programs, or activities.

If you have a documented disability and need to request academic accommodations in your courses, please refer to the SDS website at sds.umbc.edu for registration information and office procedures.

SDS email: disAbility@umbc.edu

SDS phone: 410-455-2459

If you will be using SDS approved accommodations in this class, please contact the instructor to discuss implementation of the accommodations. During remote instruction requirements due to COVID, communication and flexibility will be essential for success.

Sexual Assault, Sexual Harassment, and Gender Based Violence and Discrimination

[UMBC Policy](#) in addition to federal and state law (to include Title IX) prohibits discrimination and harassment on the basis of sex, sexual orientation, and gender identity in University programs and activities. Any student who is impacted by sexual harassment, sexual assault, domestic violence, dating violence, stalking, sexual exploitation, gender discrimination, pregnancy discrimination, gender-based harassment, or related retaliation should contact the University’s Title IX Coordinator to make a report and/or access support and resources. The Title IX Coordinator can be reached at ecr@umbc.edu or 410-455-1717.

You can access support and resources even if you do not want to take any further action. You will not be forced to file a formal complaint or police report. Please be aware that the University may

take action on its own if essential to protect the safety of the community.

If you are interested in making a report, please use the [Online Reporting/Referral Form](#). Please note that, if you report anonymously, the University's ability to respond will be limited.

Notice that Faculty and Teaching Assistants are Mandated Reporters with Mandatory Reporting Obligations

All faculty members and teaching assistants are considered Mandated Reporters, per UMBC's [Interim Policy on Sex Discrimination, Sex-Based Harassment, and Sexual Misconduct](#). Faculty and teaching assistants therefore required to report all known information regarding alleged conduct that may be a violation of the Policy to the Title IX Coordinator, even if a student discloses an experience that occurred before attending UMBC and/or an incident that only involves people not affiliated with UMBC. Reports are required regardless of the amount of detail provided and even in instances where support has already been offered or received.

While faculty members want to encourage you to share information related to your life experiences through discussion and written work, students should understand that faculty are required to report past and present sexual harassment, sexual assault, domestic and dating violence, stalking, and gender discrimination that is shared with them to the Title IX Coordinator so that the University can inform students of their [rights, resources, and support](#). While you are encouraged to do so, you are not obligated to respond to outreach conducted as a result of a report to the Title IX Coordinator. If you need to speak with someone in confidence, who does not have an obligation to report to the Title IX Coordinator, UMBC has a number of [Confidential Resources](#) available to support you:

- [Retriever Integrated Health \(Main Campus\)](#): 410-455-2472; Monday - Friday 8:30 a.m. - 5 p.m.; For After-Hours Support, Call 988.
- [Center for Counseling and Well-Being](#) (Shady Grove Campus): 301-738-6273; Monday - Thursday 10:00a.m. - 7:00 p.m. and Friday 10:00 a.m. - 2:00 p.m. (virtual) [Online Appointment Request Form](#)
- Pastoral Counseling via [The Gathering Space for Spiritual Well-Being](#): 410-455-6795; i3b@umbc.edu; Monday - Friday 8:00 a.m. - 10:00 p.m.
- [Women, Gender, and Equity Center](#) (open to students of all genders): 410-455-2714; womenscenter@umbc.edu; Monday - Thursday 9:30 a.m. - 5:00 p.m. and Friday 10:00 a.m. - 4 p.m.

Other Resources [Shady Grove Student Resources](#), [Maryland Resources](#), [National Resources](#).

Child Abuse and Neglect Please note that Maryland law and [UMBC policy](#) require that faculty report all disclosures or suspicions of child abuse or neglect to the Department of Social Services and/or the police even if the person who experienced the abuse or neglect is now over 18.

Pregnant and Parenting Students

UMBC's [Interim Policy on Sex Discrimination, Sex-Based Harassment, and Sexual Misconduct](#) expressly prohibits all forms of discrimination and harassment on the basis of sex, including pregnancy. Resources for pregnant, parenting and breastfeeding students are available through the University's [Office of Equity and Civil Rights](#). Pregnant and parenting students are encouraged to contact the Title IX Coordinator to discuss plans and ensure ongoing access to their academic program with respect to a leave of absence – returning following leave, or any other accommodation that may be needed related to pregnancy, childbirth, adoption, breastfeeding, and/or the early months of parenting. In addition, students who are pregnant and have an impairment related to their pregnancy that qualifies as disability under the ADA may be entitled to accommodations through the [Office of Student Disability Services](#).

Religious Observances & Accommodations

UMBC [Policy](#) provides that students should not be penalized because of observances of their religious beliefs, and that students shall be given an opportunity, whenever feasible, to make up within a reasonable time any academic assignment that is missed due to individual participation in religious observances. It is the responsibility of the student to inform the instructor of any intended absences or requested modifications for religious observances in advance, and as early as possible. For questions or guidance regarding religious observances and accommodations, please contact the Office of Equity and Civil Rights at ecr@umbc.edu.

Hate, Bias, Discrimination and Harassment

UMBC values safety, cultural and ethnic diversity, social responsibility, lifelong learning, equity, and civic engagement.

Consistent with these principles, [UMBC Policy](#) prohibits discrimination and harassment in its educational programs and activities or with respect to employment terms and conditions based on race, creed, color, religion, sex, gender, pregnancy, ancestry, age, gender identity or expression, national origin, veterans status, marital status, sexual orientation, physical or mental disability, or genetic information.

Students (and faculty and staff) who experience discrimination, harassment, hate, or bias based upon a protected status or who have such matters reported to them should use the [online reporting/referral form](#) to report discrimination, hate, or bias incidents. You may report incidents that happen to you anonymously. Please note that, if you report anonymously, the University's ability to respond may be limited.